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SUPERSEDING
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MILITARY SPECIFICATION

QUADRANT, ELEVATION, M12

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments

1. SCOPE

1.1 This specification covers one type of quadrant for use with Artillery motorized matériel.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS AND DRAWINGS

2.1 Specifications.—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATIONS

NN-B-601—Boxes, Wood-Cleated Plywood.

NN-B-621—Boxes, Nailed and Lock Corner.

UU-C-843—Cushioning Materials, Cellulosic.

MILITARY SPECIFICATIONS

JAN-P-100—Packaging and Packing for Overseas Shipment—General Specification.

JAN-P-106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

JAN-P-116—Packaging and Packing for Overseas Shipment — Preservation, Methods of.

JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier Materials, Waterproof, Flexible.

JAN-P-127—Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water-Resistant.

JAN-P-140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

U. S. ARMY SPECIFICATIONS

51-7—Instruments, Fire-Control; General Specification Governing the Manufacture, and Inspection of.

100-2—Standard Specification for Marking Shipments by Contractors.

(Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Drawings.—The following drawings and drawings shown thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

ARMY ORDNANCE DEPARTMENT DRAWINGS

Class 92, Division 35, Drawing 1, Quadrant, Elevation, M12.

D7693860—Fixture, Impact, Testing.

B7633839—Cleaning, Preserving, Packaging (Methods O, I and IA), Packing and Marking of Fire Control Matériel Spare Parts for Domestic Shipment and Storage, and Export Shipment.

B7633841—Packaging Instructions for Domestic Shipment and Storage, and Overseas Shipment of Aiming Post Lights and Instrument Lights.

(Copies of Army Ordnance Department Drawings may be obtained from the procuring agency or as directed by that agency. Both the title and identifying number should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 General.—All applicable requirements and tests contained in Specification 51-7 shall apply.

3.2 Material.—Material shall be in accordance with the applicable specifications and drawings.

3.3 Accuracy.—The quadrant shall meet the following requirements when set up on a suit-

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able testing fixture (mounted in a position as shown on the assembled views drawing with the mounting surfaces of the bracket in horizontal and vertical planes, respectively), with the level bubbles central, and with the indexes opposite their respective zero's (3 for angle of site):

3.3.1 *Angle of site mechanism.*—The cross-level bubble shall not be displaced more than ¼ division from the central position when the angle of site mechanism is rotated from "3" to "0" and from "3" to "6" respectively and releveled by means of the testing fixture.

3.3.2 *Elevation mechanism.*—The cross-level bubble shall not be displaced more than one division from the central position when the quadrant is elevated 800 mils and releveled by means of the testing fixture.

3.3.3 *Cross-level mechanism.*—The elevation bubble shall not be displaced more than one division from central position when the quadrant is inclined in the cross-level plane 11 degrees either right or left, and releveled by means of the cross-level adjustment knob.

3.3.4 *Backlash.*—Backlash in the elevation and angle of site mechanism shall not exceed 0.5 mil each. Backlash in the crosslevel mechanism shall not exceed 1.0 mil.

3.3.5 *Error.*—Error in the angle of site mechanism, at any setting from "3" to "6" and from "3" to "0", or the error in the elevation mechanism shall not exceed 1.5 mils, including backlash.

3.4 Temperature. — The temperature requirements shall be as specified in Specification 51–7 for type III instruments except that the high operating temperature shall be 125° F. plus maximum solar radiation.

3.5 Shock.—After being subjected to ten 150 foot pound shocks, the mount shall not be damaged and shall meet the requirements of 3.3.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection.—Unless otherwise specified, each quadrant and spare part shall be inspected for conformance with this specification and other applicable specifications, and shall be subject to inspection during process of manufacture and assembly as specified in Specification 51–7.

4.2 Tests.

4.2.1 *Accuracy.*—Each quadrant shall be tested for accuracy of elevation at not less than 6 settings and for angle of site at not less than 4 settings.

4.2.2 *Temperature.*—The temperature test shall be in accordance with type III instruments of Specification 51–7, except the temperature specified in 3.4 shall apply.

4.2.3 *Shock.*—This test shall be made on an impact testing fixture in general accordance with Drawing D7693860. The quadrant shall be mounted to simulate the "on carriage" position and the direction of the impact shall simulate the shock received by the quadrant due to the normal recoil of the gun.

5. PREPARATION FOR DELIVERY

5.1 Preservation.—Unprotected metal surfaces shall be cleaned by method C–5 (scrub with a brush or wipe with saturated cloths) and preserved with type P5 preservative (apply with a brush) as specified in JAN–P–116. Preservative shall not be applied on non-ferrous metal surfaces from which the preservative would be difficult to remove.

5.2 Packaging.—The quadrant and equipment shall be prepackaged in separate containers. The quadrant shall be packaged in accordance with Method I (in greaseproof barrier wrap or bag) of Specification JAN–P–116. The instrument light shall be packaged in accordance with Drawing B7633841. Spare lamp bulbs shall be cushioned with cellulose wadding conforming to the applicable requirements of Specification UU–C–843 and packaged in accordance with method 1A–8 of Specification JAN–P–116. A spare lamp bulb package shall be secured to each instrument light package.

5.2.1 The prepackaged items shall be packed together in a container and overwrapped with waterproof barrier material conforming with the applicable requirements of Specification JAN–P–125. The seams and flaps of the overwrap shall be sealed with water-resistant adhesive conforming to Specification JAN–P–140 and with pressure sensitive tape conforming to

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Specification JAN–P–127, Type I, II or III, Grade A or B. Packaging materials and containers shall conform to the general requirements of Specification JAN–P–100.

5.2.2 *Interior marking.*—Each container and water-resistant overwrap shall have the following identification data printed or marked in a legible manner either directly on the container and overwrap or on a label attached to the container and overwrap:

QUANTITY _____
NOMENCLATURE _____
ORD DWG. NO. _____
STOCK NO. _____
SERIAL NO.[1] _____
MFGD BY _____
DATE PACKED__ Month__ Year__

[1] Applicable to elevation quadrant and elevation quadrant with equipment packages only.

5.3 Packing.

5.3.1 *For domestic shipment.*—Elevation quadrants, with equipment, packaged as described in 5.1 shall be packed in substantial wood containers conforming to the applicable requirements of Specifications NN–B–601 or NN–B–621. Case liners are not required.

5.3.2 *For overseas shipment.*—Elevation quadrants, with equipment packaged as described in 5.1 shall be packed in a nailed wood box conforming to Style 2 or 4 of Specification JAN–P–106. Each box shall contain a water-resistant bag or case liner in which the contents are packed, fabricated from material conforming to the applicable requirements of Specification JAN–P–125. Joints and closure shall be completely sealed with water-resistant adhesive conforming with the applicable requirements of Specification JAN–P–140 in such a manner as to provide water resistance equal to the body of the material.

5.3.3 The contents of both domestic and export shipping containers shall be immobilized.

5.4 Marking.—Shipping containers, in addition to any special markings required by the contract or order, shall be marked in accordance with Specification 100–2.

5.5 Spare parts.—Packaging, packing, and marking of spare parts shall be in accordance with Drawing B7633839, and supplementary instructions supplied by the contracting officer.

6. NOTES

6.1 Ordering data.—Invitation for bids and contracts or orders should specify the following:

a. Title, number, and date of this specification.
b. Whether domestic or overseas shipment is required. (See 5.3.)

Notice.—This specification, together with specifications and drawings pertaining to it and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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